Group Coursework Submission Form

Specialist Masters Programme

Please list all names of group members: (Surname, first name) 1. Jhalak Aggarwal 2. Jiaxuan Mao 3. Abdus Sattar Mojumder	4. Aaron Neil Pinto **GROUP NUMBER:** **7**
MSc in: Actuarial Science	
Module Code: SMM540	
Module Title: Research Methods for Actuarial Professionals	
Lecturer: Rosalba Radice	**Submission Date: 01/11/2024**

Declaration:

By submitting this work, we declare that this work is entirely our own except those parts duly identified and referenced in my submission. It complies with any specified word limits and the requirements and regulations detailed in the coursework instructions and any other relevant programme and module documentation. In submitting this work we acknowledge that we have read and understood the regulations and code regarding academic misconduct, including that relating to plagiarism, as specified in the Programme Handbook. We also acknowledge that this work will be subject to a variety of checks for academic misconduct.

We acknowledge that work submitted late without a granted extension will be subject to penalties, as outlined in the Programme Handbook. Penalties will be applied for a maximum of five days lateness, after which a mark of zero will be awarded.

Marker's Comments (if not being marked on-line):

Deduction for Late Submission of assignment:

For Students:
Once marked please refer to Moodle for your final coursework grade, including your Peer Assessment grade.

Group Assignment 1_Act Sci-Group 7

Jhalak Aggarwal, Jiaxuan Mao, Abdus Sattar Mojumder, Aaron Neil Pinto

2024-10-31

Q1

Preparing Data and Determining the Number of Factors

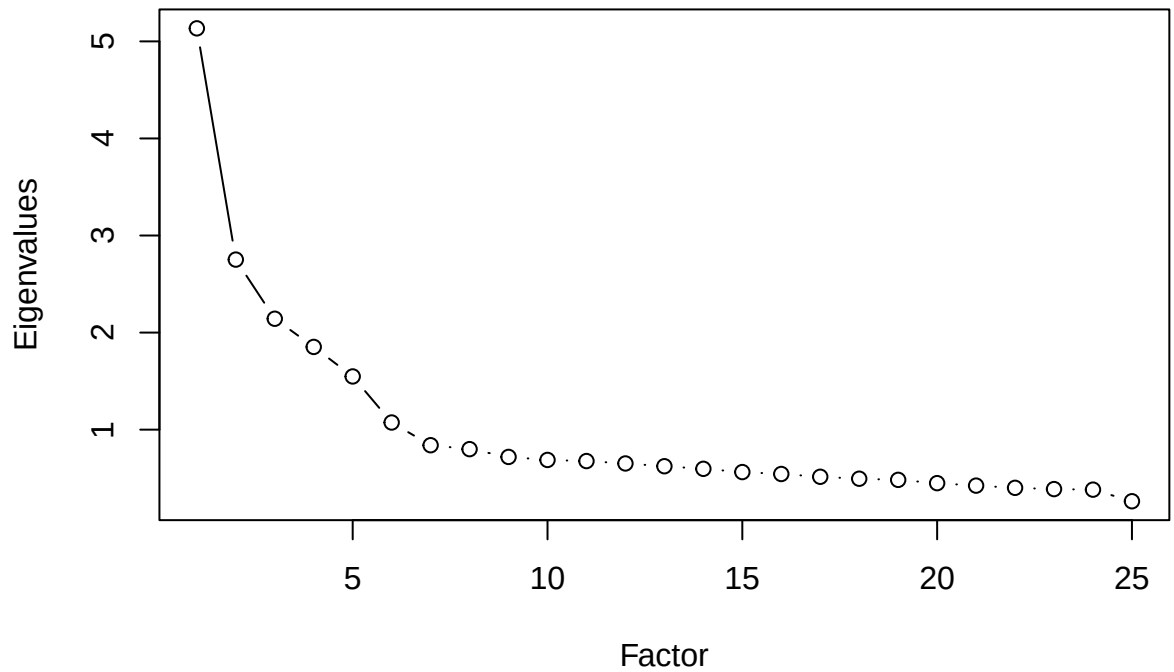
```
data<-read.table("/Users/maojiaxuan/desktop/dataset1.txt")
# Check for missing values
sum(is.na(data))
```

```
## [1] 508
```

```
# Removes rows with any NA values
data <- na.omit(data)
fa.cor<-cor(data) # correlation matrix
fa.eigen<-eigen(fa.cor)
#Deciding number of factors
number_of_factors<-length(which(fa.eigen$values>1))
number_of_factors
```

```
## [1] 6
```

```
plot(fa.eigen$values,type="b",ylab="Eigenvalues",xlab="Factor")
```



To select the number of factors, I first examined the eigenvalues, as they reflect the variance explained by each factor. Using the eigenvalue criterion (retaining factors with eigenvalues > 1), we identified 6 factors. Cattell's Scree Plot method supported this, showing a distinct elbow at the sixth factor. However, the sixth eigenvalue barely exceeds 1 (by only 0.07).

Factor Analysis

```
fa.res_1<-factanal(x=data, factors=number_of_factors, rotation="promax")
fa.res_1
```

```
##
## Call:
## factanal(x = data, factors = number_of_factors, rotation = "promax")
##
## Uniquenesses:
##    X1    X2    X3    X4    X5    X6    X7    X8    X9   X10   X11   X12   X13
## 0.675 0.482 0.473 0.696 0.516 0.637 0.498 0.685 0.424 0.566 0.613 0.453 0.522
##   X14   X15   X16   X17   X18   X19   X20   X21   X22   X23   X24   X25
## 0.432 0.599 0.273 0.302 0.480 0.504 0.652 0.663 0.703 0.510 0.756 0.640
##
## Loadings:
##     Factor1 Factor2 Factor3 Factor4 Factor5 Factor6
## X1   0.167                  -0.605           0.188
## X2   0.109                   0.679
```

```
## X3   0.262                   0.555           0.157
## X4   0.158           0.195   0.349  -0.180
## X5   0.377  -0.189           0.359           0.285
## X6                   0.601           0.128
## X7                   0.749
## X8                   0.607                  -0.185
## X9                  -0.700  -0.105           0.569
## X10 -0.107   0.100  -0.571                   0.330
## X11 -0.651  -0.207   0.180
## X12 -0.732
## X13  0.560                           0.233   0.358
## X14  0.724                          -0.133   0.242
## X15  0.488   0.223   0.232           0.177
## X16  0.147   0.936                          -0.122
## X17          0.941                          -0.196
## X18          0.649                           0.102
## X19 -0.369   0.350                           0.212
## X20 -0.167   0.370           0.121  -0.171   0.161
## X21  0.171                           0.477   0.247
## X22                                 -0.502   0.164
## X23  0.272                           0.568   0.293
## X24 -0.273                   0.145   0.348   0.198
## X25                                 -0.581   0.165
##
##                Factor1 Factor2 Factor3 Factor4 Factor5 Factor6
## SS loadings      2.732   2.615   2.267   1.492   1.487   1.187
## Proportion Var   0.109   0.105   0.091   0.060   0.059   0.047
## Cumulative Var   0.109   0.214   0.305   0.364   0.424   0.471
##
## Factor Correlations:
##         Factor1 Factor2 Factor3 Factor4 Factor5 Factor6
## Factor1   1.000 -0.3773 -0.3327 -0.1848  -0.326  0.1603
## Factor2  -0.377  1.0000  0.0747  0.0754   0.114 -0.4859
## Factor3  -0.333  0.0747  1.0000  0.2033   0.302  0.2775
## Factor4  -0.185  0.0754  0.2033  1.0000   0.102 -0.0214
## Factor5  -0.326  0.1136  0.3019  0.1016   1.000  0.2177
## Factor6   0.160 -0.4859  0.2775 -0.0214   0.218  1.0000
##
## Test of the hypothesis that 6 factors are sufficient.
## The chi square statistic is 896.7 on 165 degrees of freedom.
## The p-value is 5.73e-101
```

```r
library(psych)
# Perform factor analysis with 6 factors and obtain RMSEA
fa_result <- fa(data, nfactors = 6, fm = "ml")
```

```
## Loading required namespace: GPArotation
```

```r
print(fa_result$RMSEA[1])
```

```
##      RMSEA
## 0.04266438
```

After fitting the model with 6 factors, the chi-square test yielded a significant p-value, indicating that the 6-factor solution may not sufficiently explain the data. To address this, I tested models with additional factors, finding that the p-value became insignificant only at 13 factors. However, retaining 13 factors risks overfitting and complicates interpretation.
I then considered the RMSEA (Root Mean Square Error of Approximation), which is less sensitive to large sample sizes and often provides a more practical fit assessment. With a good fit typically indicated by RMSEA values below 0.08, the 6-factor model achieves this, suggesting it may be adequate.
I also applied factor rotation to enhance interpretability by maximizing high loadings and minimizing low ones on each factor while preserving communalities and uniqueness.

Interpretation of the result

```r
print(fa.res_1$loadings, cut=0.2)
```

```
##
## Loadings:
##     Factor1 Factor2 Factor3 Factor4 Factor5 Factor6
## X1                          -0.605
## X2                           0.679
## X3   0.262                   0.555
## X4                           0.349
## X5   0.377                   0.359           0.285
## X6                   0.601
## X7                   0.749
## X8                   0.607
## X9                  -0.700                   0.569
## X10                 -0.571                   0.330
## X11 -0.651  -0.207
## X12 -0.732
## X13  0.560                           0.233   0.358
## X14  0.724                                   0.242
## X15  0.488   0.223   0.232
## X16          0.936
## X17          0.941
## X18          0.649
## X19 -0.369   0.350                           0.212
## X20          0.370
## X21                                  0.477   0.247
## X22                                 -0.502
## X23  0.272                           0.568   0.293
## X24 -0.273                           0.348
## X25                                 -0.581
##
##                Factor1 Factor2 Factor3 Factor4 Factor5 Factor6
## SS loadings      2.732   2.615   2.267   1.492   1.487   1.187
## Proportion Var   0.109   0.105   0.091   0.060   0.059   0.047
## Cumulative Var   0.109   0.214   0.305   0.364   0.424   0.471
```

```r
fa_result_2 <- fa(data, nfactors = 5, fm = "ml")
print(fa_result_2$RMSEA[1])
```

```
##      RMSEA
## 0.05382273
```

By examining significant loadings, we can assign names to each factor. Factor 1 is characterized by high loadings on items like "Know how to captivate people," "Make friends easily," and "Take charge," with negative loadings on "Don't talk a lot" and "Find it difficult to approach others." These traits collectively suggest Sociability. Similarly, Factor 2 aligns with Emotional Instability; Factor 3 reflects Conscientiousness (a common term in personality psychology capturing diligence, responsibility, and organization); Factor 4 represents Empathy; and Factor 5 embodies Intellectual Curiosity.
For Factor 6, the loadings are weak and too dispersed to summarize meaningfully. Additionally, the RMSEA for the 5-factor model is 0.054, well below 0.08, indicating a good fit. Thus, a 5-factor solution should provide a balanced and interpretable model for this dataset.

Factor Scores

```r
fa.res_2<-factanal(x=data, factors=number_of_factors-1, rotation="promax",scores="Bartlett")
factor_names<-c("Sociability","Emotional Instability","Conscientiousness","Empathy","Intellectual Curio
colnames(fa.res_2$scores) <- factor_names
head(fa.res_2$scores)
```

```
##       Sociability Emotional Instability Conscientiousness     Empathy
## 61617 -0.29155616            0.34256227       -1.59272281 -1.25048844
## 61618  0.03519056            0.56935979       -0.76430747 -0.16703695
## 61620  0.71811479            0.24402844       -0.04079361 -0.82245279
## 61621 -0.11913339            0.01418467       -1.38046290  0.02362399
## 61622 -0.40819005            0.69894853       -0.08147229 -1.01525748
## 61623  0.24306810            1.36211765        1.77469309  0.21556714
##       Intellectual Curiosity
## 61617             -2.3255007
## 61618             -0.2788221
## 61620              0.3138904
## 61621             -1.4657324
## 61622             -0.9964153
## 61623              0.6582918
```

Factor scroes of latent factor for each participant can be used in further analysis.
In conclusion, the 25 self-reported personality measurements in this dataset can be effectively condensed into 5 factors: Sociability, Emotional Instability, Conscientiousness, Empathy, and Intellectual Curiosity. Together, these five factors explain 41.4% of the total variance. The factor scores enable us to observe each participant's standing on these latent traits, making it easier to interpret the dataset and identify meaningful patterns.

Q2

Introduction:

This analysis uses Multidimensional Scaling (MDS) to explore the purchasing habits of 15 shoppers based on how often they buy items across nine product categories. The data set, taken from a text file named Dataset2.txt, includes data on the number of items purchased by each shopper for products like laptops, smartphones, headphones, Cameras, and gaming consoles.

Importing Library and Data from a text file.

```r
# Package for MDS functions
library(MASS)

# Importing dataset from a text file
data_set<-read.table("/Users/maojiaxuan/desktop/dataset2.txt",header = TRUE)
```

We would also check for missing data(NA) for the given dataset. This step is essential to ensure data integrity.

```r
# Check for missing values
if (sum(is.na(data_set)) > 0) {
  cat("Missing values found in the dataset")
} else {
  cat("No missing values in the dataset.\n")
}
```

```
## No missing values in the dataset.
```

The summary() provides a quick review of the central tendencies and spread of the data, and apply() which calculates variance for each column, showing that Smartphones, Headphones, and Tablets have high variances.

```r
summary(data_set[,-1])              #summary of data set
```

```
##    Smartphone       Headphones       Smartwatch        Tablet         Camera
##  Min.   : 1.000   Min.   : 0.000   Min.   :0.000   Min.   : 0.0   Min.   :0.0
##  1st Qu.: 2.000   1st Qu.: 1.500   1st Qu.:1.000   1st Qu.: 1.0   1st Qu.:0.5
##  Median : 4.000   Median : 3.000   Median :3.000   Median : 3.0   Median :1.0
##  Mean   : 5.067   Mean   : 3.867   Mean   :2.733   Mean   : 4.2   Mean   :2.2
##  3rd Qu.: 8.000   3rd Qu.: 5.500   3rd Qu.:3.000   3rd Qu.: 7.0   3rd Qu.:4.0
##  Max.   :13.000   Max.   :12.000   Max.   :8.000   Max.   :13.0   Max.   :6.0
##     Monitor         Keyboard          Mouse       Gaming.Console
##  Min.   :0.000   Min.   : 2.000   Min.   :0.000   Min.   : 0.0
##  1st Qu.:1.000   1st Qu.: 3.000   1st Qu.:1.000   1st Qu.: 1.5
##  Median :2.000   Median : 6.000   Median :2.000   Median : 4.0
##  Mean   :2.267   Mean   : 5.333   Mean   :1.933   Mean   : 3.6
##  3rd Qu.:3.500   3rd Qu.: 7.000   3rd Qu.:3.000   3rd Qu.: 5.0
##  Max.   :6.000   Max.   :10.000   Max.   :4.000   Max.   :10.0
```

```r
apply(data_set[,-1], 2, var)       #Calculate variance for each column
```

```
##     Smartphone     Headphones     Smartwatch         Tablet         Camera
##      12.495238      11.266667       4.780952      13.600000       4.457143
##        Monitor       Keyboard          Mouse Gaming.Console
##       3.209524       6.809524       1.780952       6.971429
```

Distance Calculation

Next, we calculate the distance matrix using the Euclidean method. This matrix measures the dissimilarities between the 15 shoppers based on their purchasing patterns.

```
# Calculating the distance using the Euclidean method
dist_mat<- dist(data_set, method = "euclidean")
```

Multidimensional Scaling (calculating distance and performing MDS and stress analytics)

We intend to use Multidimensional scaling on a data set to visualize high-dimensional data in two-dimensional (k=2) using cmdscale() and isoMDS() functions. Additionally, we have performed Stress Analysis to find the best-fit approach for the data representation. Lower stress values indicate a more accurate representation of the data set in 2D space. We can see that the stress value for the classical MDS approach is 0.0494 approximately closer to zero, indicating more accuracy than the non-metric approach which has a stress level of 6.2922. This suggests that classical MDS is more suitable for our dataset.

```
# Perform Classical (Metric) MDS
c_mds<- cmdscale(dist_mat, k=2, eig=TRUE)
# Perform Non-metric MDS using iosMDS from MASS package
no_mds<- isoMDS(dist_mat, k=2)
```

```
## initial  value 9.372025
## iter   5 value 6.426909
## final  value 6.292154
## converged
```

```
# stress analysis
stress<- sum((dist_mat- dist(c_mds$points))^2)/sum(dist_mat^2)

print(paste("Stress value for classical approach MDS:",round(stress, 4)))
```

```
## [1] "Stress value for classical approach MDS: 0.0494"
```

```
print(paste("Stress value for non-metric MDS:",round(no_mds$stress, 4)))
```

```
## [1] "Stress value for non-metric MDS: 6.2922"
```

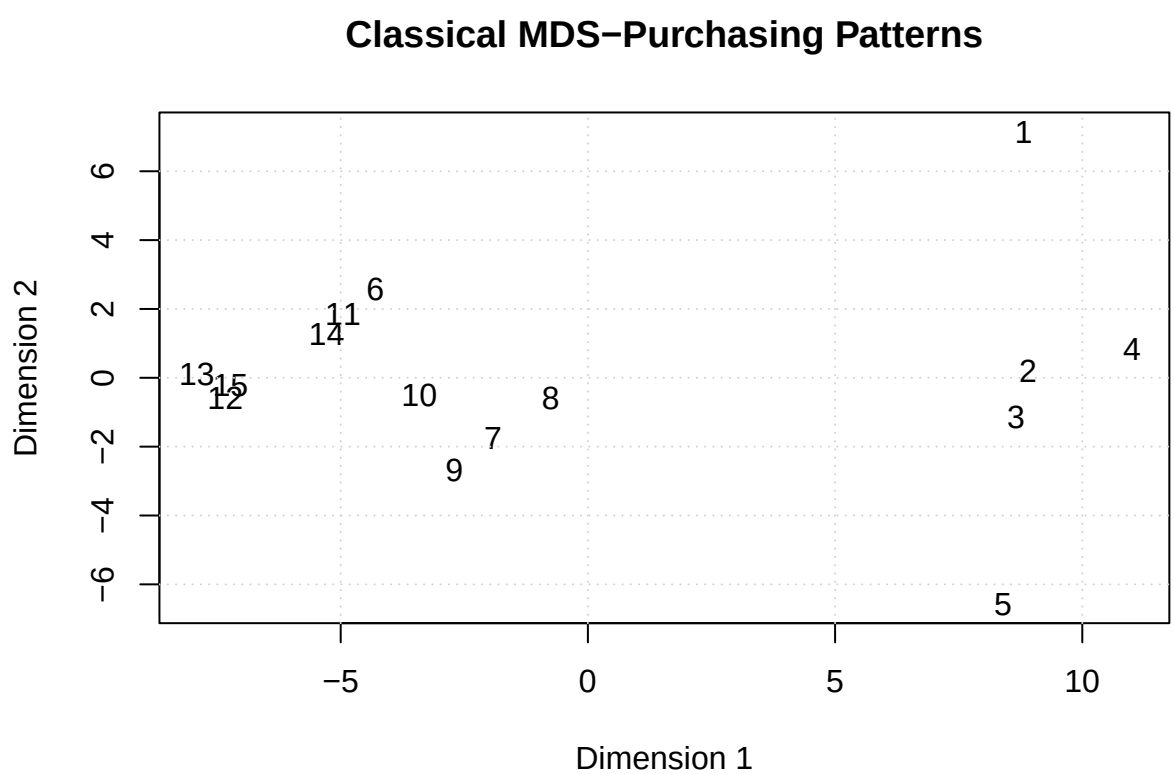
Ploting the classical MDS and Visualization

The coordinates for two-dimensional plots by using cmdscale(). We use plot() to plot the graph and also label the points, that represent individual shoppers.

```
# coordinates in two dimensions for plot
mds_coords <- cmdscale(dist_mat, k=2)
print(mds_coords)
```

```
##                  [,1]       [,2]
## Shopper 1   8.8168338  7.1584888
## Shopper 2   8.9046373  0.2181726
## Shopper 3   8.6604012 -1.1308700
## Shopper 4  11.0047058  0.8403321
## Shopper 5   8.3967625 -6.5785227
## Shopper 6  -4.2986817  2.5755355
## Shopper 7  -1.9178638 -1.7581493
## Shopper 8  -0.7537833 -0.5857052
## Shopper 9  -2.7017280 -2.6932306
## Shopper 10 -3.4089394 -0.4939563
## Shopper 11 -4.9509014  1.8718628
## Shopper 12 -7.3310728 -0.5867593
## Shopper 13 -7.9095519  0.1037512
## Shopper 14 -5.2815964  1.2666011
## Shopper 15 -7.2292219 -0.2075507
```

```r
#plotting the graph in two-dimensional space
plot(mds_coords, type = "n", xlab = "Dimension 1", ylab = "Dimension 2",
     main = "Classical MDS-Purchasing Patterns")
text(mds_coords,labels = 1:nrow(data_set), cex=1)

# adds grid lines to the plot
grid()
```



Result interpretation:

The plot generated from Classical MDS shows distinct clusters among shoppers, indicating that those with similar purchasing habits are grouped closely together. For example, Shopper 1, Shopper 2, and Shopper 3 are clustered in one area of the plot, suggesting they tend to purchase similar products. In contrast, Shopper 6, which has a markedly different purchasing pattern, is positioned further away from this group.

The variance calculations support these findings. The higher variances for smartphones (12.495238) and tablets (13.600000) indicate diverse consumer preferences, suggesting that different marketing strategies could be applied to capture consumer interest in these categories. On the other hand, the lower variance in gaming consoles and monitors indicates more uniform purchasing patterns among shoppers.

Conclusion:

In summary, this Multidimensional scaling is highly effective for revealing purchasing trends among consumers. Using this classical MDS approach we can provide a clear visualization of shopper relationships, which illustrates the clusters that can guide the marketing strategies. By understanding these trends, retailers can better cater to customer preferences and enhance their engagement efforts, ultimately driving sales. The findings highlight the necessity of customizing marketing efforts based on observed purchasing behaviors in the plot, especially in categories exhibiting high variability. This targeted methodology using MDS has the potential to yield more effective marketing outcomes and strengthen connections with consumers.

Q3

Introduction

We are performing a principle component analysis (PCA) on the dataset so that we can understand major patterns in consumer behaviour based of different indicators that are mentioned in the dataset such as income and age.

```r
# Load the dataset
data <- read.table("/Users/maojiaxuan/desktop/dataset3.txt",header= TRUE, sep = " ",dec = "." )

# Check structure and summary of the dataset
str(data)
```

```
## 'data.frame':    500 obs. of  11 variables:
##  $ Income                     : num  70564 41530 55447 59493 56064 ...
##  $ Age                        : num  57.9 20.9 56.5 43.3 41.5 ...
##  $ Spending_Score             : num  2.66 29.17 8.29 28.28 16.7 ...
##  $ Shopping_Frequency         : int  6 7 8 8 6 10 12 5 14 6 ...
##  $ Brand_Loyalty              : num  5.85 6.48 8.75 10.01 5.67 ...
##  $ Website_Visits             : int  27 28 38 28 29 29 33 24 33 30 ...
##  $ Social_Media_Engagement    : num  54.6 47 48.6 53.4 55.3 ...
##  $ Email_Open_Rate            : num  24.2 37.8 10.2 33.7 37.6 ...
##  $ Customer_Satisfaction_Score: num  104.6 62.7 98.6 70.9 68.5 ...
##  $ Promotions_Utilized        : int  3 5 0 4 2 3 2 1 0 0 ...
##  $ Cart_Abandonment_Rate      : num  58.82 28.89 1.25 54.71 26.02 ...
```

```r
summary(data)
```

```
##      Income           Age        Spending_Score   Shopping_Frequency
##  Min.   : 5104   Min.   :18.06   Min.   :-12.73   Min.   : 0.000
##  1st Qu.:40091   1st Qu.:29.49   1st Qu.: 12.14   1st Qu.: 6.000
##  Median :49430   Median :40.85   Median : 24.54   Median : 7.000
##  Mean   :49549   Mean   :40.95   Mean   : 25.37   Mean   : 7.486
##  3rd Qu.:59534   3rd Qu.:52.58   3rd Qu.: 35.92   3rd Qu.: 9.000
##  Max.   :94488   Max.   :64.97   Max.   : 75.24   Max.   :17.000
##  Brand_Loyalty    Website_Visits  Social_Media_Engagement Email_Open_Rate
##  Min.   :-0.702   Min.   :14.00   Min.   :17.60           Min.   : 0.1232
##  1st Qu.: 5.717   1st Qu.:27.00   1st Qu.:42.87           1st Qu.:26.4278
##  Median : 7.911   Median :30.00   Median :49.56           Median :54.2274
##  Mean   : 8.103   Mean   :30.35   Mean   :49.70           Mean   :50.9998
##  3rd Qu.:10.478   3rd Qu.:34.00   3rd Qu.:56.46           3rd Qu.:75.9438
##  Max.   :19.995   Max.   :49.00   Max.   :83.29           Max.   :99.8852
##  Customer_Satisfaction_Score Promotions_Utilized Cart_Abandonment_Rate
##  Min.   : 28.17              Min.   :0.000       Min.   : 0.4422
##  1st Qu.: 63.30              1st Qu.:1.000       1st Qu.:27.3155
##  Median : 73.53              Median :2.000       Median :53.1603
##  Mean   : 73.81              Mean   :2.524       Mean   :51.1590
##  3rd Qu.: 83.41              3rd Qu.:4.000       3rd Qu.:74.6845
##  Max.   :124.57              Max.   :5.000       Max.   :99.9680
```

We can observe that consumer behavior is determined by 11 variables in the this dataset.

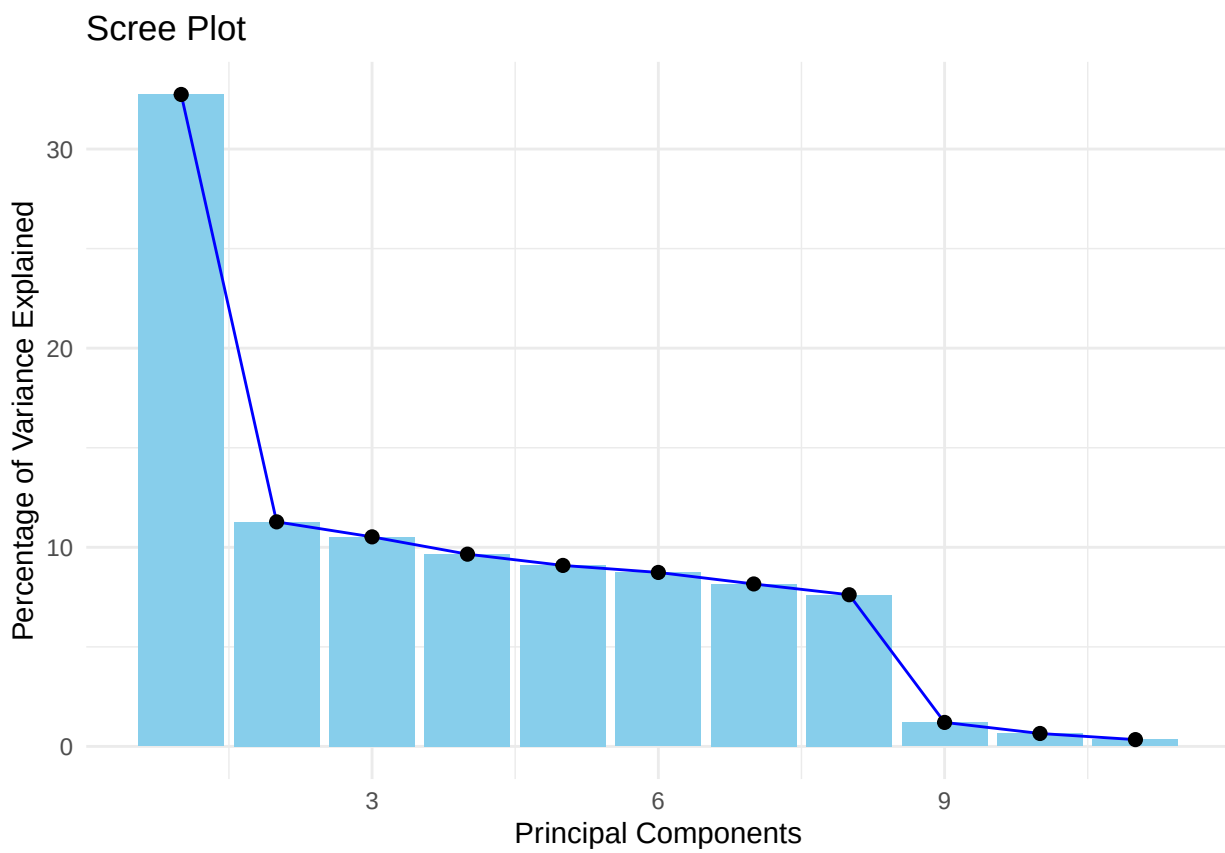
```r
# Check for missing values
if (sum(is.na(data)) > 0) {
  cat("Missing values found in the dataset. Consider handling them before proceeding.\n")
} else {
  cat("No missing values in the dataset.\n")
}
```

```
## No missing values in the dataset.
```

```r
# Standardize the data (scaling to mean 0 and variance 1)
data_scaled <- scale(data)

# Perform PCA
pca <- prcomp(data_scaled, center = TRUE, scale. = TRUE)

# Summary of PCA to observe variance explained by each component
print(summary(pca))
```

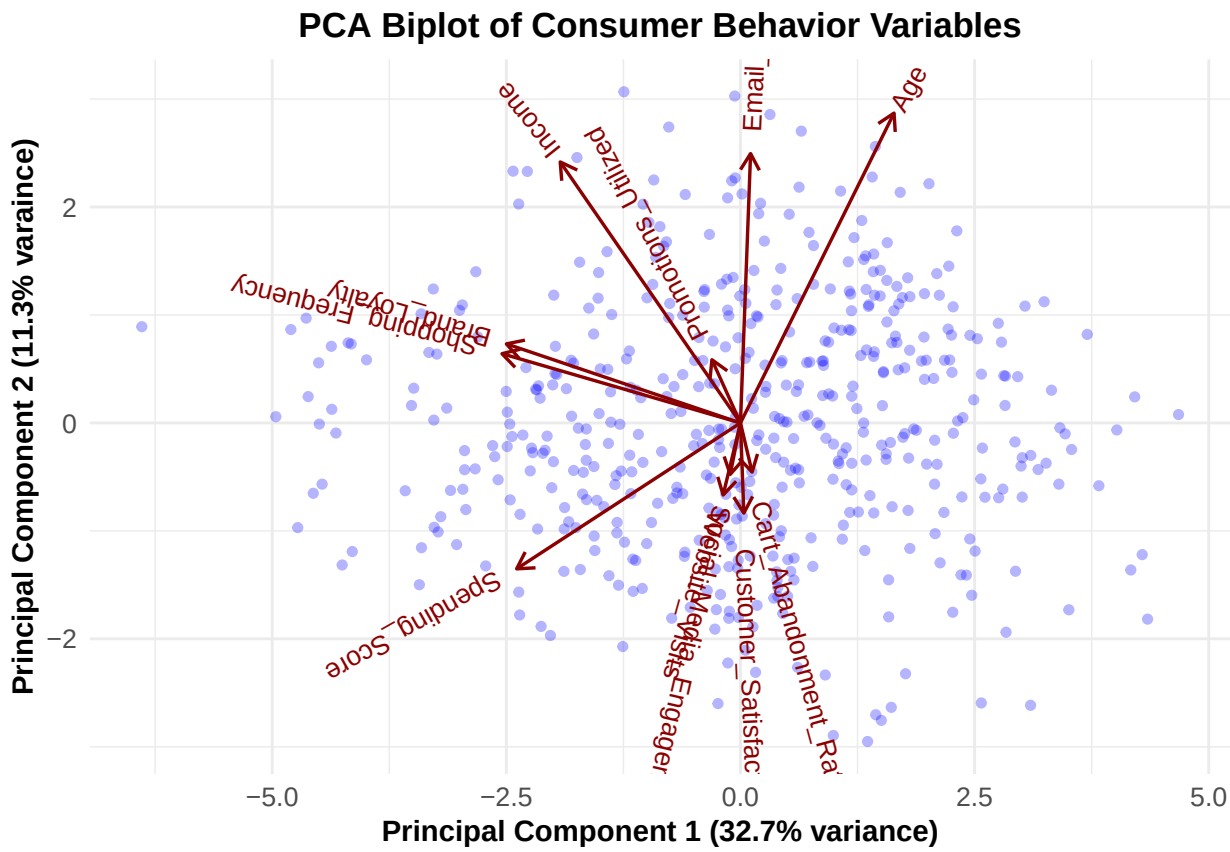
```
## Importance of components:
##                          PC1    PC2    PC3     PC4    PC5     PC6    PC7
## Standard deviation     1.8978 1.1138 1.0759 1.03053 0.9999 0.98035 0.9474
## Proportion of Variance 0.3274 0.1128 0.1052 0.09654 0.0909 0.08737 0.0816
## Cumulative Proportion  0.3274 0.4402 0.5454 0.64198 0.7329 0.82025 0.9019
##                           PC8     PC9   PC10    PC11
## Standard deviation     0.91536 0.36414 0.2674 0.19412
## Proportion of Variance 0.07617 0.01205 0.0065 0.00343
## Cumulative Proportion  0.97802 0.99007 0.9966 1.00000
```

```r
# Scree plot
pca_var <- pca$sdev^2
pca_var_percent <- pca_var / sum(pca_var) * 100
scree_data <- data.frame(PC = 1:length(pca_var_percent), Variance = pca_var_percent)

ggplot(scree_data, aes(x = PC, y = Variance)) +
  geom_bar(stat = "identity", fill = "skyblue") +
  geom_line(color = "blue") +
  geom_point(size = 2) +
  ggtitle("Scree Plot") +
  xlab("Principal Components") +
  ylab("Percentage of Variance Explained") +
  theme_minimal()
```



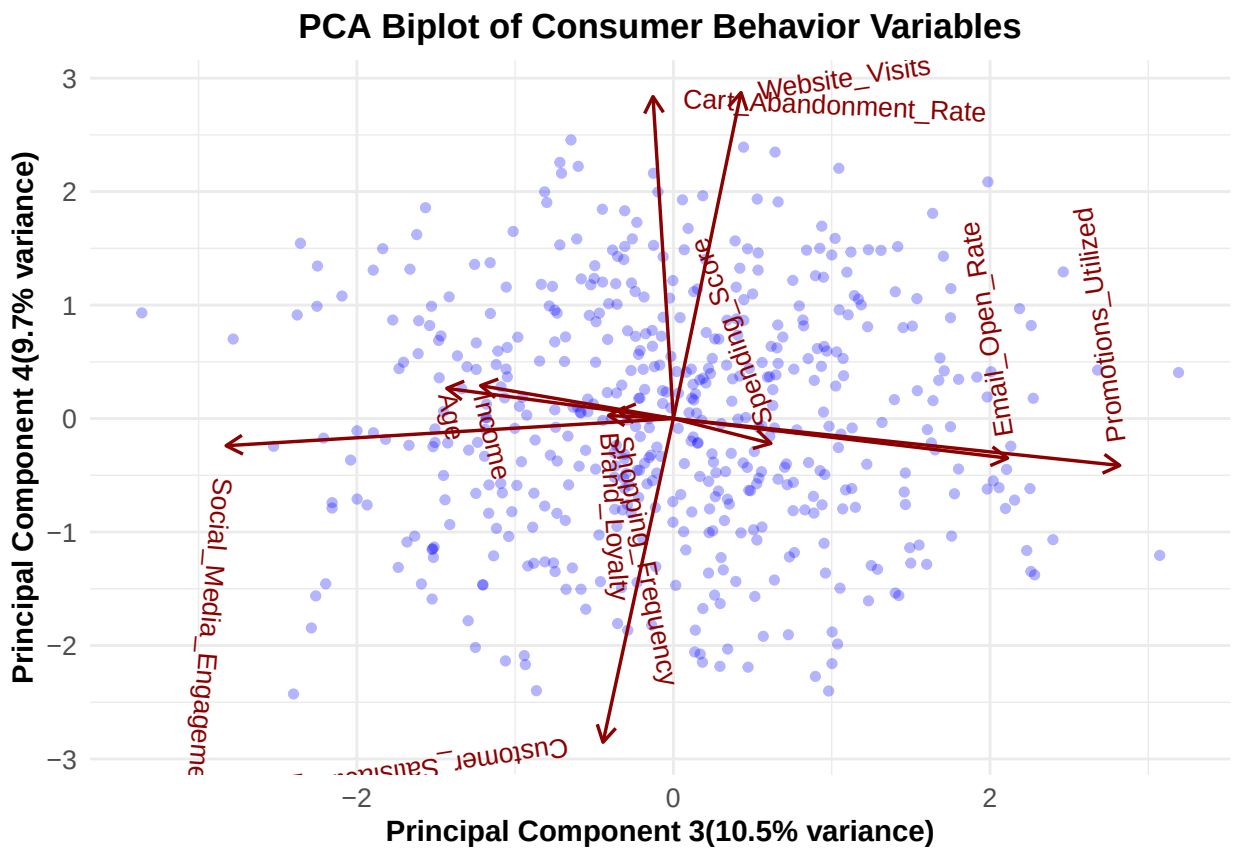

```r
## Biplot for the the first two principal components
# Extract PCA scores (observations) and loadings (variables)
scores <- data.frame(pca$x[, 1:2])
loadings <- data.frame(pca$rotation[, 1:2])
loadings$var <- rownames(loadings)

# Calculate angles for labels to match the direction of arrows
loadings$angle <- atan2(loadings$PC2, loadings$PC1) * 180 / pi

# Set the scale factor for arrows and labels
arrow_scale <- 5
```

```r
# Create a biplot
ggplot() +
  geom_point(data = scores, aes(x = PC1, y = PC2), color = "blue", alpha = 0.3, size = 1.3) +
  geom_segment(data = loadings, aes(x = 0, y = 0, xend = PC1 * arrow_scale, yend = PC2 * arrow_scale),
               arrow = arrow(length = unit(0.25, "cm")), color = "darkred", linewidth = 0.6) +
  geom_text(data = loadings, aes(x = PC1 * arrow_scale, y = PC2 * arrow_scale, label = var, angle = angl
            color = "darkred", size = 3.5, hjust = -0.1, vjust = 0.5) +
  labs(title = "PCA Biplot of Consumer Behavior Variables",
       x = "Principal Component 1 (32.7% variance)",
       y = "Principal Component 2 (11.3% varaince)") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(face = "bold"),
    axis.text = element_text(size = 10)
  )
```




```r
##Biplot for the the third and fourth principal components
# Extract PCA scores (observations) and loadings (variables)
scores_2 <- data.frame(pca$x[, 3:4])
loadings_2 <- data.frame(pca$rotation[, 3:4])
loadings_2$var <- rownames(loadings_2)

# Calculate angles for labels to match the direction of arrows
loadings_2$angle <- atan2(loadings_2$PC3, loadings_2$PC4) * 180 / pi
```

```r
# Set the scale factor for arrows and labels
arrow_scale_2 <- 5

# Create a biplot
ggplot() +
  geom_point(data = scores_2, aes(x = PC3, y = PC4), color = "blue", alpha = 0.3, size = 1.3) +
  geom_segment(data = loadings_2, aes(x = 0, y = 0, xend = PC3 * arrow_scale, yend = PC4 * arrow_scale)
               arrow = arrow(length = unit(0.25, "cm")), color = "darkred", linewidth = 0.6) +
  geom_text(data = loadings_2, aes(x = PC3 * arrow_scale, y = PC4 * arrow_scale, label = var, angle = ar
            color = "darkred", size = 3.5, hjust = -0.1, vjust = 0.5) +
  labs(title = "PCA Biplot of Consumer Behavior Variables",
       x = "Principal Component 3(10.5% variance)",
       y = "Principal Component 4(9.7% variance)") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(face = "bold"),
    axis.text = element_text(size = 10)
  )
```



Observations

1. The scree plot is showing the variance explained by the principal components. It can be observed that the variance is in a descending order where the first 8 principal components have a cumulative

variance of around 97%.

2. By interpreting the scree plot, we can conclude that the first 4 - 5 principal component will be able to provide a useful summary of our dataset As they are accounting for approximately 73% of the variance. but for our analysis we are using the 1 & 2 and the 3 & 4 principal component.

3. Bi plot of the first 2 principal component: we can observe that spending score, shoppinf frequency and brand loyalty are the primary influencers of PC1. they are suggesting that if a customers shops more often then their brand loyalty is higher as well as their spending scores. Whereas, PC2 is predominantly influenced by Age, email open rate and income suggesting how different demographich and their income affects their behavior as consumers.

4. Bi plot of the third and fourth principal component: PC3 is affected by promotion utilization and social media engagement depicting the influence of media on the consumer behavior but as they are in the opposite direction it suggests they consumers chooses either one but not both in normal case scenarios. in the bi plot PC4 is majorly affected by customer satisfaction score, website visits and cart abondonment rate indicating that cart abandonment rate is usually work in the opposite direction of customer satisfaction score but alongside website visits.

Conclusion

The scree plot helps select the number of components to retain like is in this case the first 4 prinicipal, while the biplot provides a visual interpretation of variable influence and observation patterns in the consumer behaviour. Together, these insights help us in understanding complex relationships in consumer behavior data that is provided by our dataset.